

Project Report: Nike Sales Performance Analysis

1. Overview of the Project

This project's goals were to evaluate regional performance, identify important profitability drivers, and comprehend how pricing strategies (discounts) affect the bottom line by analyzing Nike's sales data.

Important Tools Used:

For data cleaning and exploratory data analysis (EDA), I used Python (Pandas, Matplotlib, Seaborn).

SQL: For extracting business logic and conducting structured queries.

Power BI: For visual storytelling and interactive dashboarding.

2. Data Cleaning and Preparation

The Jupyter notebook addressed a number of inconsistencies found in the raw dataset (Nike_Sales_Uncleaned.csv):

Size: Imputed using the Mode for Missing Value Treatment.

Units Sold: grouped by Product_Line and imputed using the median.

MRP: Managed missing values to guarantee precise revenue estimates.

Feature Engineering: To classify profit into High Profit, Profit, and Lower Profit, a margin_type column was created.

Standardized regional names (such as combining "Hyd" and "Hyderabad" or "Bengaluru" and "Bangalore") are an example of data normalization.

3. Important Performance Indicators (KPIs)

The following core metrics were set up based on the analysis:

Total Revenue: The total amount of money made from sales in all areas.

Average Profitability: The average profit for each order.

Project Report: Nike Sales Performance Analysis

Order Distribution: A look at how many transactions happen through different sales channels.

Product Performance: Found the five products that made the most money, such as Air Jordan and Mercurial Superfly.

4. Strategic Insights

A. Performance by region

Delhi, Mumbai, Bangalore, Hyderabad, and Pune are some of the most important places for sales.

The SQL analysis found the specific areas that brought in the most total revenue, which made it possible to market to those areas more effectively.

B. Analysis of the Product Line

Running, soccer, lifestyle, training, and basketball are some of the product lines.

The study found that the Average MRP varies between lines, with Basketball and Lifestyle often having higher prices.

C. Distribution of Sales Channels

There are two ways to place orders: in stores and online.

The project looked at the number of orders in each channel to see which one works better for certain types of products.

D. The "Discount Trap" Insight:

A key finding from the SQL analysis was that items with discounts of more than 40% often lost money.

This means that high discounts do bring in more business but less profit.

Project Report: Nike Sales Performance Analysis

5. Visual Highlights (Power BI and Matplotlib)

The presentation has pictures for:

Profitability Heatmap: It shows the null distribution and data density.

Revenue by Region: Bar charts that show which areas are the most important.

Profit vs. Product Line: Finding out which groups are "Cash Cows" and which are "Underperformers."

Margin Distribution: A pie chart or histogram that shows how many orders are high profit and how many are low profit.

6. Conclusion & Recommendations

Recommendations:

Rethink Discounting: Analyze the discounting policy for low, margin products. Avoid giving discounts beyond 30% so as not to fall into the loss zone.

Concentrate on High, Margin Products: Air Jordan, Phantom GT, and other high, margin products should be given priority in inventory and marketing budget.

Regional Strategy: Focus on improving logistics and retail network in major revenue cities like Delhi and Mumbai.

Online Growth: Take advantage of online bundles in the "Lifestyle" segment, which attracts a high volume of digital consumers.

Project Report: Nike Sales Performance Analysis

Courtesy:

- Python: <https://www.python.org/>
- Pandas: <https://pandas.pydata.org/>
- Matplotlib: <https://matplotlib.org/>
- Seaborn: <https://seaborn.pydata.org/>
- Sqlalchemy: <https://www.sqlalchemy.org/>
- Postgresql: <https://www.postgresql.org/>
- Data Source: <https://www.kaggle.com/datasets/nayakganesh007/nike-sales-uncleaned-dataset>
- Company name used: <https://www.nike.com/in/>